

CSE46901: Introduction to Robotics

Simulated Milestone 2

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Introduction

Welcome to CSE46901. We have prepared a few milestones before the final project, in which you will use a physical robot to execute a mission given by us. The following milestones will give you an idea how to interact with [ROBOTIS' Turtlebot3 waffle Pi](#) with a [Manipulator Arm](#) using the Robot Operating System.

For more information on ROS itself, this free book from the robot's manufacturer will help you be familiar with the Robot Operating System.

Official: <https://www.robotis.us/ros-robot-programming-book-digital-copy/>

Mirror: https://ai.unist.ac.kr/~chiu/robot/books/ROS_Robot_Programming_EN.pdf

Alternatively, following wiki's instructions will be helpful to you. Start from "Navigating the ROS Filesystem" after installing ROS based on the instructions we will provide you below.

<http://wiki.ros.org/ROS/Tutorials>

Submission requirement

In accordance with the online classes only policy for March, we have changed the first milestone to be done in simulation. In this simulated milestone, you will demonstrate your ability to follow basic instructions in setting up a programming environment for ROS.

We are not trying to differentiate students' performance with milestones. The only grade that will be given is Pass/Fail. However, we do expect you to provide us with evidence that you have followed through the milestones by posting a video of each group following through with the tasks in Part 1 and 2. Requirements of what to be included in the video will be provided below.

Once your group is done with the video demonstration, please submit it on the blackboard. Each group will submit one video. However, each member will be asked to take parts in the video in specified ways to prevent free-riding.

Once we verify that you have submitted the milestone requirement, you will be awarded Pass/Fail. You can resubmit videos to challenge your Fail status until the final project time period so that all groups can eventually pass the milestone requirement with no penalty for resubmission. (i.e. no difference between teams who get it right the first try vs teams who retries). However, if we resume offline lectures, there would be logistical limit of each teams being able to physically interact with the turtlebot.

Instruction

Part1

This tutorial is based on the following tutorial. However, because turtlebot officially does not support ROS Noetic on ubuntu 20.0.4 for simulating the manipulator, do not blindly follow robotis' tutorial. However, you may still head there for more information.

<https://emanual.robotis.com/docs/en/platform/turtlebot3/manipulation/>

This tutorial assumes that you have followed setup progress in milestone 1

On your own Computer:

1) Set up Open-Manipulator-X:

- Use a root account to install a few more ROS packages according to https://emanual.robotis.com/docs/en/platform/openmanipulator_x/ros_setup/#install-ubuntu-on-pc

```
sudo apt-get update && sudo apt-get upgrade
```

```
sudo apt-get install ros-noetic-ros-controllers ros-noetic-gazebo*  
ros-noetic-moveit*
```

```
sudo apt-get install ros-noetic-industrial-*
```

- Compile following pertaining to open manipulator from source

```
cd ~/catkin_ws/src/  
git clone https://github.com/ROBOTIS-GIT/DynamixelSDK.git  
git clone https://github.com/ROBOTIS-GIT/dynamixel-workbench.git  
git clone https://github.com/ROBOTIS-GIT/dynamixel-workbench-msgs.git  
git clone https://github.com/ROBOTIS-GIT/open_manipulator.git  
git clone https://github.com/ROBOTIS-GIT/open_manipulator_msgs.git  
git clone https://github.com/ROBOTIS-GIT/open_manipulator_simulations.git  
git clone https://github.com/ROBOTIS-GIT/robotis_manipulator.git  
cd ~/catkin_ws && catkin_make
```

```
source ~/catkin_ws/devel/setup.bash
```

- Compile following files pertaining to turtlebot3 from source

```
cd ~/catkin_ws/src/  
git clone https://github.com/ROBOTIS-GIT/turtlebot3_manipulation.git  
git clone  
https://github.com/ROBOTIS-GIT/turtlebot3_manipulation_simulations.git  
cd ~/catkin_ws && catkin_make  
  
source ~/catkin_ws/devel/setup.bash
```

Part2

Next we need you to demonstrate your simulated environment with the operation of a robotic manipulator on Gazebo. Two ways will be presented here: rviz and Robotis GUI.

How to run simulation using Rviz

This tutorial is based on the following tutorial. However, because turtlebot officially does not support ROS Noetic on Ubuntu 20.0.4, do not blindly follow robotis' tutorial. However, you may still head there for more information.

<https://emanual.robotis.com/docs/en/platform/turtlebot3/manipulation/>

This tutorial assumes that you have completed the setup process for remote-pc correctly on milestone 1

1) Make sure ip address setting on bashrc is still correct by typing

```
ifconfig
```

and

```
vi ~/.bashrc
```

If you changed any of the settings, type

```
source ~/.bashrc
```

2) Open simulated world with manipulator attached turtlebot3:

- On Remote-PC (each in separate terminal windows)

```
roscore
```

```
roslaunch turtlebot3_manipulation_gazebo turtlebot3_manipulation_gazebo.launch
```

- **Press play button on the gazebo window next to “Real time factor”**. If you do not do this rviz will be stuck in “initializing” state. Then on separate terminal windows,

```
roslaunch turtlebot3_manipulation_moveit_config move_group.launch
```

3) Test Open-Manipulator-X (Moveit RViz):

- On Remote-PC (each in separate terminal windows)

```
roslaunch turtlebot3_manipulation_moveit_config moveit_rviz.launch
```

- You can now move the blue ball around and click "Plan and Execute" to move the arm to the desired location.

- Try out different things within Rviz gui to get a feel of turtlebot3 with manipulator arm's capabilities.

- Note: If the arm does not move according to your command, check if "turtlebot3_manipulation_bringup" is still running. Some ros nodes essential for operation of the manipulator arm can crash without your intervention.

- Note: Under Motion planning -> Planning-> Commands box there exists a feedback message that gives feedback if turtlebot managed to successfully execute your commands. If the output is "fail" or for whatever reason the arm does not move according to your command, check if "turtlebot3_manipulation_bringup" is still running. Some ros nodes essential for operation of the manipulator arm can crash without your intervention.

How to run simulation using ROBOTIS GUI

This tutorial is based on the following tutorial. However, because turtlebot officially does not support ROS Noetic on Ubuntu 20.0.4, do not blindly follow robotis' tutorial. However, you may still head there for more information.

<https://emanual.robotis.com/docs/en/platform/turtlebot3/manipulation/>

This tutorial assumes that you have completed the setup process for remote-pc correctly on milestone 1

1) Make sure ip address setting on bashrc is still correct by typing

```
ifconfig
```

and

```
vi ~/.bashrc
```

If you changed any of the settings, type

```
source ~/.bashrc
```

2) Test Simulated world and Open-Manipulator-X:

- On Remote-PC

```
roscore
```

- On Remote-PC, open a new terminal and launch (each in separate terminal windows)

```
roslaunch turtlebot3_manipulation_gazebo turtlebot3_manipulation_gazebo.launch
```

- **Press play button on the gazebo window next to "Real time factor"**. If you do not do this, rviz will be stuck in "initializing" state. Then, each on separate terminal windows,

```
roslaunch turtlebot3_manipulation_moveit_config move_group.launch
```

```
roslaunch turtlebot3_manipulation_gui turtlebot3_manipulation_gui.launch
```

- Click "Timer Start" on the GUI

- Click "Read joint angle" on the GUI under joint space

- You can now issue commands by changing Joint 1 ~ Joint 4's angles and clicking "send" button or manipulate gripper by clicking "Gripper open" button or "Gripper close" button.

- Try out different things within the Open manipulator control gui to get a feel of turtlebot3 with manipulator arm's capabilities.

Demo requirement

You will have to submit some photos and/or videos showing that you have successfully controlled the manipulator via

- (1) rviz and
- (2) the ROBOTIS GUI Controller.

Please try to learn the user interface of rviz and the ROBOTIS GUI Controller, and demonstrate what you have learnt in the videos.

After a sequence of demos showing four things, to discourage freeriding, each member should include a clip in the video with their own voice explaining one of the following. (i.e. member a will explain 1, member b will explain 2... if your team member is below or more than 5, you can skip or re-explain answered topics)

Please explain the following topics in 30 seconds or less each. It does not have to be precise. You can google the topics and respond in your own words.

1. What is a catkin workspace?
2. Why do we source `~/catkin_ws/devel/setup.bash` ?
3. What is moveit?
4. How do you think rviz or gui is commanding the robotic arm?
5. How can you use what you have learnt to grab a bottle?